Prabhavit Gupta



AI Platform Engineer: Model Gateways, MCP Servers & Agent Orchestration

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Summary

AI Platform and Software Engineer with 3 years of production experience architecting model gateways, Model Context Protocol (MCP) servers, agent orchestration runtimes, and high-throughput RAG systems. Deep hands-on expertise building unified LLM routing layers with semantic caching and fallback mechanisms, deploying tool-using multi-step agents, and building automated evaluation harnesses with regression gates in CI/CD. Proficient in Python, TypeScript, GCP (Vertex AI), Docker, and PostgreSQL. Based in Venice, Italy (CET time zone) with an EU Long-Term Residence Permit (permanent unrestricted EU work authorization across all EU member states, no sponsorship required); available immediately for full-time remote roles.

Technical Expertise

- **AI Platform & Agent Infrastructure:** Model Context Protocol (MCP) servers, model gateways, agent orchestration, tool/function calling, LangChain, semantic routing, context window management, human-in-the-loop guardrails.
- **LLM Observability & Evaluation:** Automated evaluation harnesses (offline golden datasets, online transcript tracing), latency and cost tracking, prompt caching, semantic caching, CI regression gates.
- **Software Development & Backend Systems:** Python (FastAPI, asyncio, pandas, NumPy), TypeScript, Node.js, RESTful APIs, PostgreSQL schema design, Redis asynchronous queues, JSON schema validation.
- **RAG & Vector Retrieval:** Retrieval-Augmented Generation, vector embeddings, hybrid semantic and keyword retrieval, metadata filtering, chunking strategies, automated document ingestion pipelines.
- **DevOps & Cloud Fundamentals:** Docker containerization, GCP (Vertex AI, Cloud Run, GCS), GitOps, CI/CD pipeline automation, automated testing with Playwright, Linux system administration.
- **Languages & Authorization:** English (C1 Professional, fluent), Hindi (Native), Italian (Beginner). EU Long-Term Residence Permit (unrestricted EU work authorization across all EU member states, no sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer & AI Systems Architect *London, UK (remote)*

- Designed and operated production customer-facing AI agents and model routing pipelines for a marketplace platform serving 4,500+ active enterprise users, integrating platform catalogs with external APIs.
- Developed custom Model Context Protocol (MCP) servers and tool definitions exposing PostgreSQL data layers and external web services securely to LLM agents.
- Engineered multi-step agent reasoning chains with strict tool execution constraints, schema validation, and guardrails, ensuring zero ungrounded actions in production flows.
- Built an automated evaluation harness and regression testing pipeline measuring precision, grounding, and recall against production interaction transcripts, cutting hallucination rates.
- Implemented semantic caching and token budget enforcement with Redis, reducing repetitive LLM inference costs and p95 latency.
- Architected and maintained resilient TypeScript/Node.js backend services and PostgreSQL data layers, utilizing asynchronous Redis queues to handle high-throughput query spikes.
- Developed automated integration and end-to-end regression test suites with Playwright, verifying API contracts across continuous deployment cycles.

Urlink **Apr 2025 - Jul 2025**
Project Manager & Solutions Developer (Freelance) *Barcelona, Spain (remote)*

- Led technical discovery, solution architecture, and client-facing delivery of customized web solutions and automated data processing pipelines for European client platforms.
- Built automated data ingestion and transformation services connecting third-party APIs to PostgreSQL databases, maintaining high data freshness and integrity.
- Delivered interactive functional prototypes to validate user flows and technical requirements prior to sprint execution, accelerating cycle times.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Automation & Integration Developer *Venice, Italy*

- Built and deployed custom LLM integrations, model gateway proxies, and webhook pipelines for European clients using Python, TypeScript, and n8n.
- Replaced brittle manual data processing with automated LLM-assisted extraction pipelines featuring comprehensive error handling and fallback routing.
- Authored technical architecture specifications, API documentation, and deployment guides to support client engineering teams in operating self-hosted workflows.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous training in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.